

Data Analysis Report

Financial Loan



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COMPANY INTRODUCTION

Apex Lending Group is a fast-growing consumer lending company that provides personal loans, debt consolidation loans, and small business loans across the United States. In 2021 we issued 38,576 loans worth millions, but our default and late-payment rates are higher than industry benchmarks. Our risk team is under pressure to improve loan approval decisions, pricing strategy, and collection processes before the next funding round.

PROBLEM

- ❖ Too many loans are becoming delinquent or defaulting, especially in certain grades, purposes, and income segments.
- ❖ We want to understand which customer profiles (income, employment length, home ownership, purpose) are safest vs riskiest.
- ❖ Interest rate (int_rate) and installment amount seem to influence repayment behavior.
- ❖ We need clear insights on **Debt-to-Income (DTI)** impact, loan purpose risk, and grade vs actual performance.
- ❖ **Goal:** Reduce bad loans by 15–20% in 2026 while maintaining healthy loan volume.

KEY BUSINESS QUESTIONS

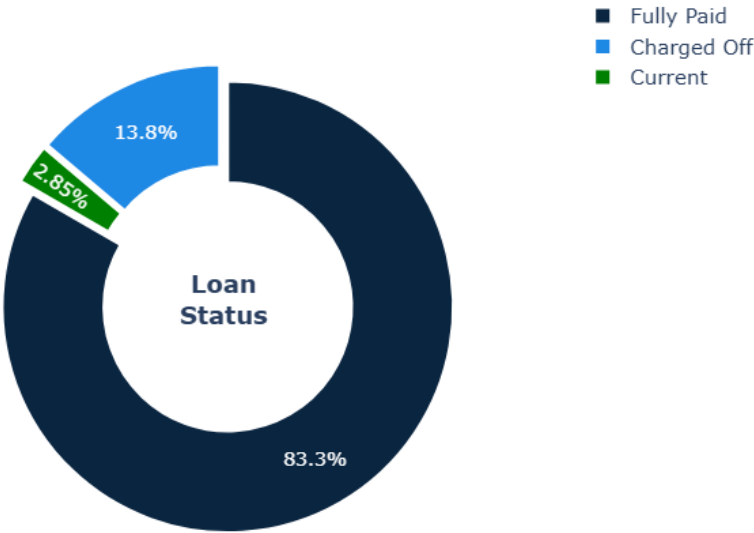
1. Overall performance
2. Customer & Risk Segmentation
3. Pricing & Repayment Behavior
4. Geographic & Behavioral Insights
5. Revenue & Risk Signals
6. Business Recommendations(Suggestions)

KEY FINDINGS

OVERALL PERFORMANCE

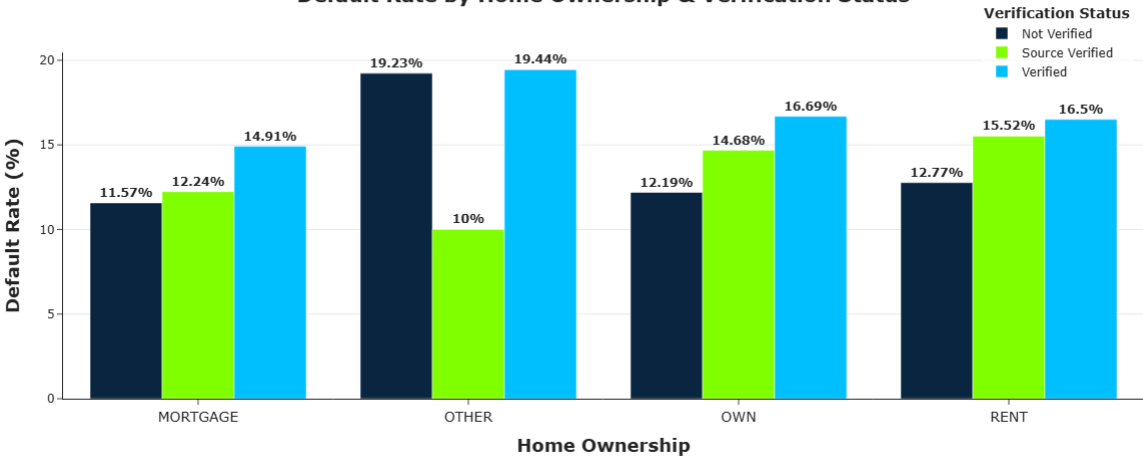
Total Loans Issued	38,576	Total Amount Lent	\$435,757,075
Average Loan Size	\$11,296.07	Average Interest Rate	12.05%
Total Payment Received	\$473,070,933	Overall Default Rate	13.82 %

Loan Status Distribution



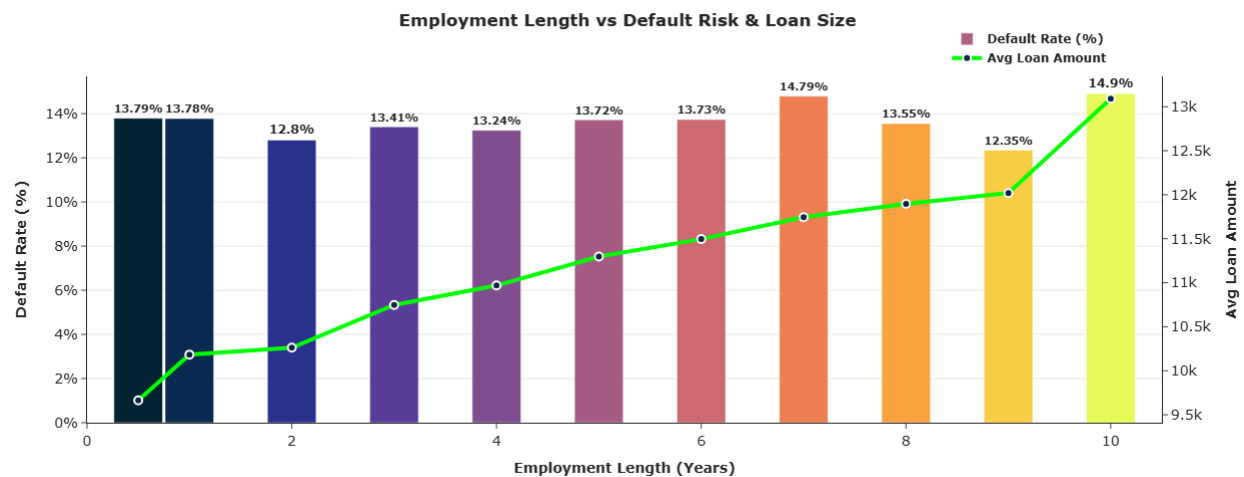
CUSTOMER & RISK SEGMENTATION

Default Rate by Home Ownership & Verification Status

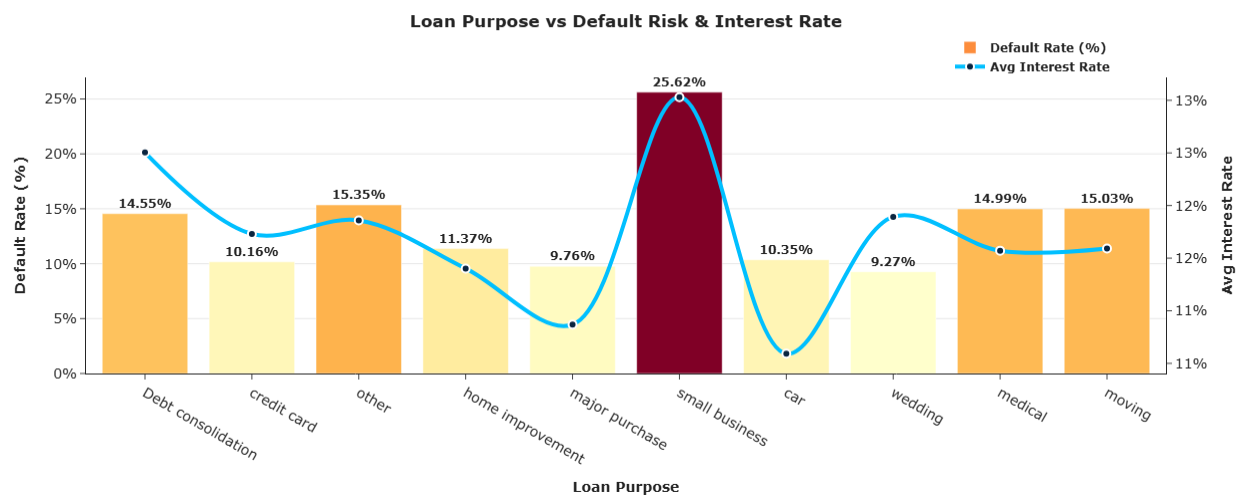


Income	Default Rate
0 - 30k	17.3%
30k - 60k	15.3%
60k - 100k	12.1%
100k - 200k	10.4%
200k +	10.2%

- Higher the Income lower the default rate.



- Higher the years, higher the **average loan amount**.



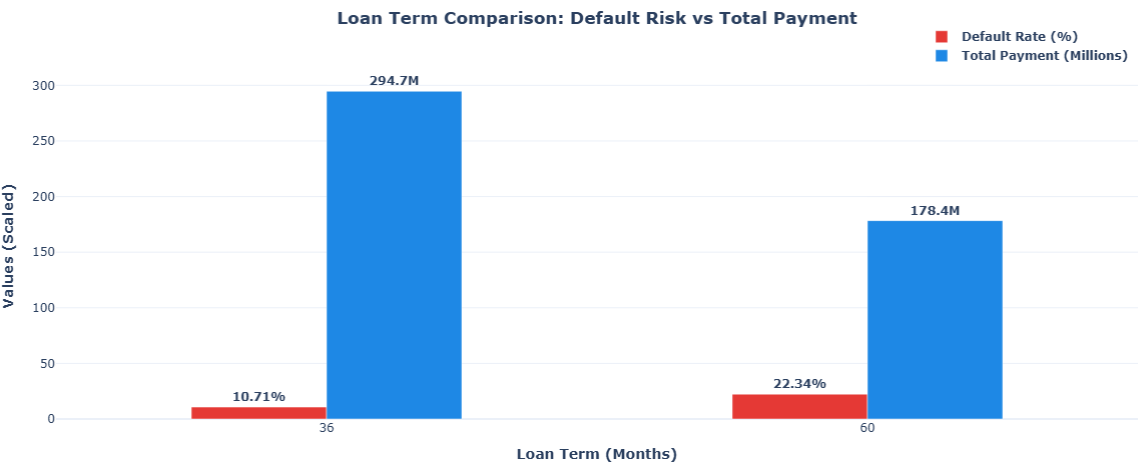
- Highest loan purpose is **starting small business**.
- Lowest loan purpose is for **car**.

PRICING & REPAYMENT BEHAVIOR

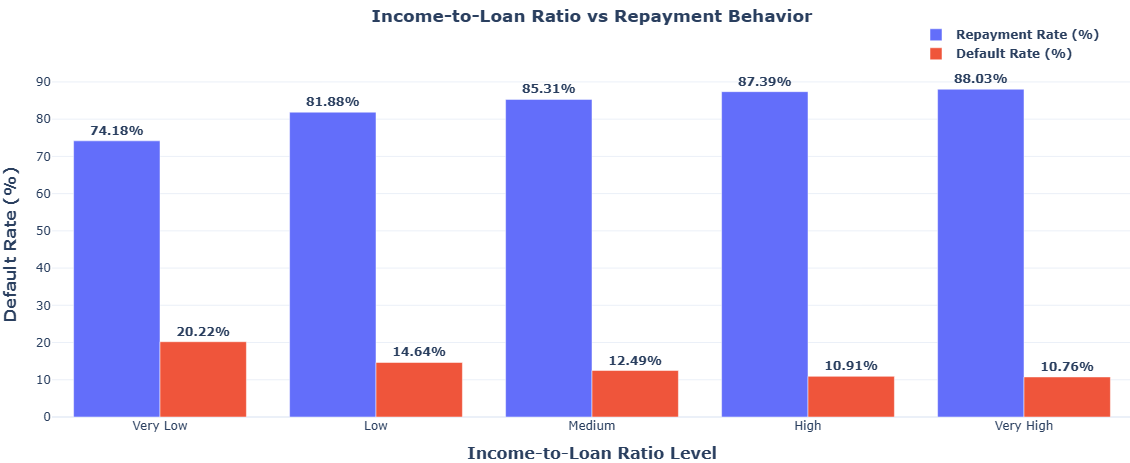
Grade	Interest Rate
A	7.5%
B	11%
C	13.5%
D	15.7%
E	17.7%
F	19.7%
G	21.4%

- Higher the grade, higher the **interest rate**.

CORRELATION BETWEEN INSTALLMENT & DEFAULT: 0.027

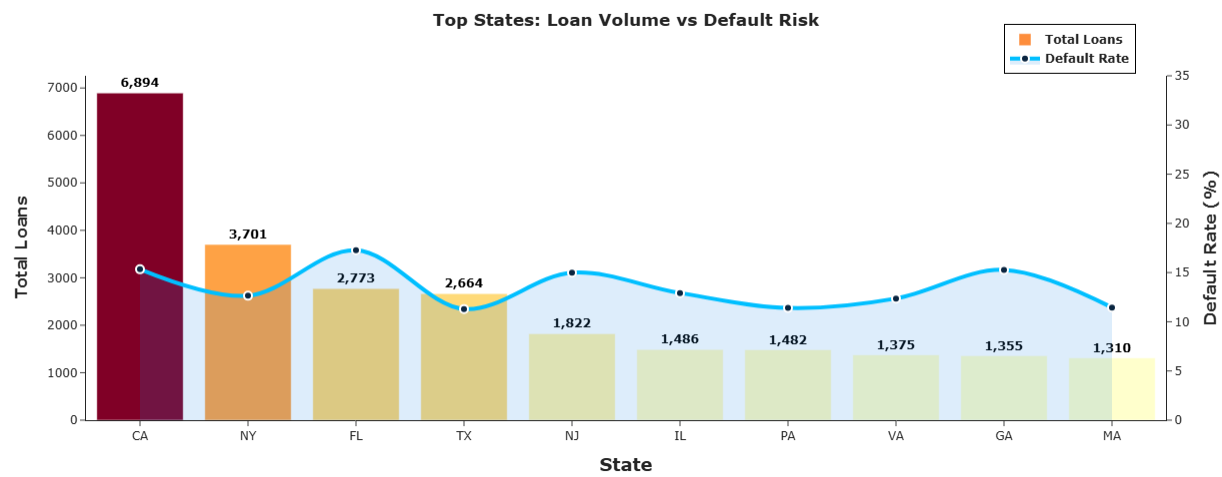


- 36 Months terms has higher Payment than 60 Months term.
- 60 Months term has higher default rate than 36 Months term.



GEOGRAPHIC & BEHAVIORAL INSIGHTS

Countries	Number of Loans	Countries	Number of Loans
CA	6894	IL	1486
NY	3701	PA	1482
FL	2773	VA	1375
TX	2664	GA	1355
NJ	1822	MA	1310

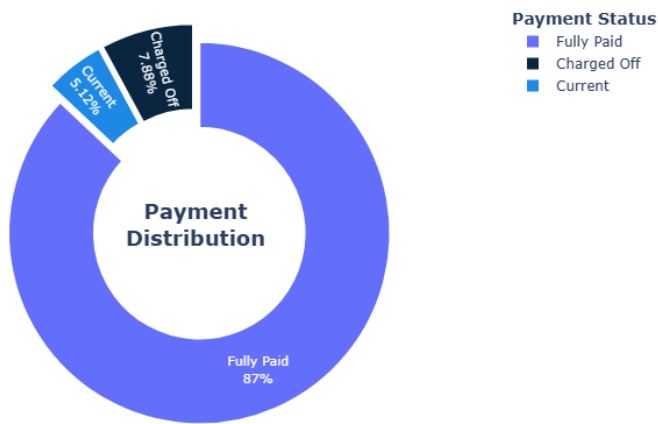


1. REVENUE & RISK SIGNALS

- Revenue only from Fully Paid loans: **411,586,256\$**

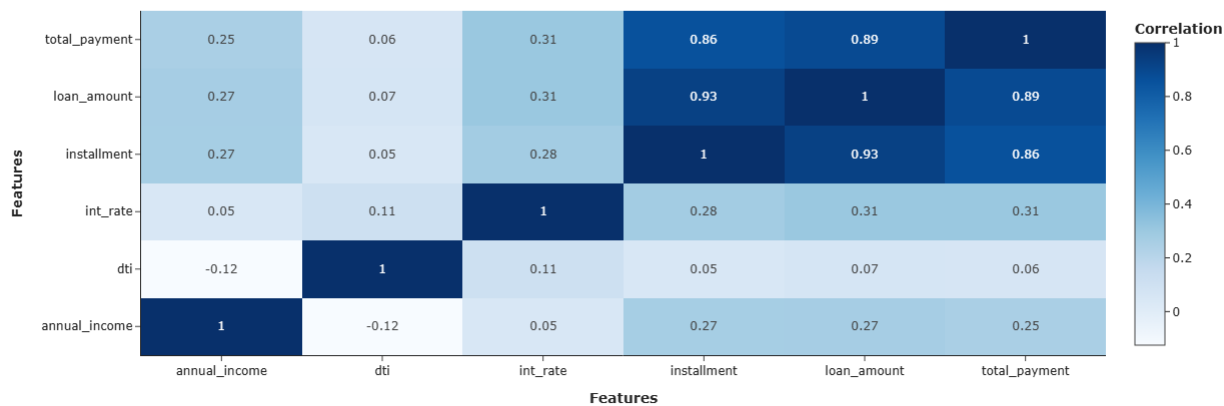
Loan Status	Total Loans	Total Payment	Payment Share	Total Loss
Charged Off	5333	\$37,284,763	7.8%	\$28,247,462
Current	1098	\$24,199,914	5.1%	0
Fully Paid	32145	\$411,586,256	87%	0

Total Payment by Loan Status



CORRELATION

Correlation Heatmap: Financial Features



BUSINESS RECOMMENDATIONS(SUGGESTIONS)

- 1. Reduce exposure to high-risk loan grades (**E–G**) as they have significantly higher default rates. Focus more on safer grades like **A** and **B**.
- 2. Limit long-term loans (**60 months**) since they show much higher default rates compared to **36-month** loans.
- 3. Strengthen **credit policies** for high-risk purposes such as **small business loans**, which have the highest default rates.

4. Encourage lending to higher-income customers, as default risk decreases significantly with increasing income levels.
5. Apply stricter screening for borrowers with high debt-to-income (DTI) ratios, especially in the **15–25%** range where risk peaks.
6. Promote loans with **higher income-to-loan ratios**, as they show better repayment performance and lower default rates.
7. Improve **monitoring** of high installment loans, as default risk increases with higher installment amounts.
8. Maintain risk-based pricing strategy, as interest rates are effectively aligned with loan risk across grades.
9. Focus **marketing** and **lending** in lower default regions (**e.g., Texas, Pennsylvania**) and review strategies in high-risk states like **Florida**.
10. Minimize losses from charged-off loans by **improving early risk detection** and **strengthening collection strategies**.